The School of Mathematics

Density-Aware Graph Geodesic Stress Testing on a Learned Macroeconomic Manifold

by

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Abstract

[abstract]

Acknowledgments

[acknowledgments]

Own Work Declaration

[own work declaration]

AI disclosure

[see guidance document]

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1 Introduction

James thinks the intro might have too many sections. Maybe not all necessary.

1.1 motivation

Limitations of linear interpolation through Euclidean space; it gives unrealistic combinations of variables.

Incoporate JDKF into motivation - argument runs that Baker has diffusion maps but linearly interpolates; can we find density aware paths

Each month gives many observations, but these could be represented using far fewer dimensions.

- learn low-dimensional representation of macro-financial states;
- construct stress transitions that respect the learned geometry and historical density;
- determine whether temporal regimes emerge in this latent representation.

1.2 Research questions

Main questions:

- Can diffusion maps recover a stable and interpretable low-dimensional representation of macro-financial conditions?
- Do graph-based and density-aware stress paths remain in more historically plausible regions than direct linear interpolation?

Extension questions:

- Do hidden Markov regimes fitted in diffusion space correspond to recognised economic regimes, including NBER recessions?
- Do density-aware geodesics pass through the inferred regimes in a temporally and economically coherent way?

1.3 contributions

Have done:

- Implementation and numerical validation of a diffusion-map pipeline;
- Construction of density-aware graph geodesics for macroeconomic stress testing;
- Lifting of latent stress paths into economically interpretable variables

Aim to do:

- Fitting of a Gaussian HMM to diffusion coordinates rather than the original high-dimensional observations;
- Joint evaluation of geometric paths, historical density, and temporal regimes.

2 Background

2.1 High-dimensional state vectors and the manifold hypothesis

A macroeconomic state vector is a point

$$z_t = (z_t^1, \dots, z_t^D) \subset \mathbb{R}^D$$

representing a variety of different macroeconomic variables at a time t, e.g. fuel prices, interest rates, etc. Although D may be large, the collection of observed states may concentrate near a lower-dimensional set

$$\mathcal{M} \in \mathbb{R}^D.$$

This is the manifold hypothesis. In this dissertation, the manifold is not assumed known analytically; instead, it is estimated numerically from data.

2.2 Latent diffusions and spectral coordinates

Explanation of the latent process and how the diffusion map is suitable for this model. This is heavily based on the JDKF paper.

Assumed latent process:

$$d\theta_t = -\nabla U(\theta_t), dt + \sqrt{2} dW_t,$$

and its infinitesimal generator:

$$\mathscr{L} = \Delta - \nabla U \cdot \nabla.$$

Generator eigenfunctions provide dynamically meaningful coordinates. The essential relation is

$$\psi_j(z_t) \approx \varphi_j(\theta_t)$$

where ψ_j is obtained from the finite-data graph and φ_j is an eigenfunction of the limiting differential operator.

2.3 Diffusion maps

This outlines how we actually build the diffusion maps, rather than the theory above.

Diffusion maps are a nonlinear dimensionality-reduction technique which use local similarities between observations to recover the geometry of a data set [2]. Suppose that

$$\mathcal{Z} = \{z_1, \dots, z_N\} \subset \mathbb{R}^D$$

is a collection of observations sampled from, or concentrated near, a lower-dimensional manifold

$$\mathcal{M} \subset \mathbb{R}^D.$$

The method constructs a weighted graph on the observations and uses the spectral properties of a random walk on this graph to obtain intrinsic coordinates for the data.

The first step is to calculate the pairwise squared Euclidean distances

$$\delta_{ij} = \|z_i - z_j\|^2.$$

These distances are converted into affinities using the Gaussian kernel

$$W_{ij} = \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right), \tag{2.1}$$

where the bandwidth $\varepsilon > 0$ controls the scale over which two observations are regarded as neighbours. Mention the fact that this is different to Mahalanobis distance used in [1]. Mahalanobis is not sensitive to nonlinearity and correlation. You have to estimate the correlations to use this; with $\tilde{Z} \in \mathbb{R}^{N\times D}$ this would introduce errors. Using the quantile data, we don't have to worry about the nonlinearity because it makes everything Gaussian, but we need to explain the correlation thing. The matrix

$$W = (W_{ij})_{i,j=1}^{N}$$

is symmetric and contains values between zero and one. Nearby observations have affinity close to one, whereas the affinity between distant observations is close to zero. Since each observation has zero distance from itself, $W_{ii} = 1$.

The weighted degree of observation i is defined by

$$q_i = \sum_{j=1}^{N} W_{ij}. \tag{2.2}$$

Unlike the degree of an unweighted graph, q_i does not simply count the number of adjacent vertices. It measures the total affinity of z_i to the sample. Consequently, a large value of q_i indicates that z_i has many nearby observations, or several observations to which it has strong affinity. Conversely, a value close to one indicates that almost all of the degree is contributed by the diagonal entry $W_{ii} = 1$, and hence that z_i is nearly isolated at the chosen bandwidth. The degree therefore acts as a kernel-based measure of local sampling density.

If the observations are sampled non-uniformly from $\mathcal{M}$, densely sampled regions may dominate the resulting random walk. Diffusion maps allow this effect to be controlled through the density-normalised affinity matrix

$$W_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^{\alpha} q_j^{\alpha}}, \qquad \alpha \in [0, 1]. \tag{2.3}$$

The parameter α determines how strongly the empirical sampling density is removed. The choice $\alpha = 0$ leaves the original affinities unchanged, whereas $\alpha = 1$ removes the leading influence of non-uniform sampling density and, under standard asymptotic conditions, yields an approximation to the Laplace–Beltrami operator associated with the underlying manifold [2]. Intermediate choices retain some dependence on the sampling distribution.

A second degree vector is then formed from the normalised affinities:

$$d_i^{(\alpha)} = \sum_{j=1}^{N} W_{ij}^{(\alpha)}. \tag{2.4}$$

Writing

$$D_\alpha = \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right),$$

the diffusion transition matrix is

$$P = D_\alpha^{-1} W^{(\alpha)}, \qquad P_{ij} = \frac{W_{ij}^{(\alpha)}}{\sum_{k=1}^{N} W_{ik}^{(\alpha)}}. \tag{2.5}$$

Each row of P sums to one, so P can be interpreted as the transition matrix of a Markov chain on the data. In particular, P_{ij} is the probability of moving from observation z_i to observation z_j in one step. The entries of P^t give the corresponding transition probabilities after t steps.

This random-walk interpretation induces a notion of distance that depends on the connectivity of the data rather than solely on the ambient Euclidean distance. Two observations are considered close if random walks beginning at them have similar transition probabilities after t steps. The resulting diffusion distance may be written as

$$D_t^2(z_i, z_j) = \sum_{k=1}^{N} \frac{\left(P_{ik}^t - P_{jk}^t\right)^2}{\pi_k}, \tag{2.6}$$

where π_k denotes the stationary probability associated with observation z_k. This compares the full distributions of possible destinations from the two starting points. Consequently, diffusion distance is robust to individual short-circuit connections and reflects the presence of multiple paths through the data cloud.

Although P is generally not symmetric, it is similar to the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}. \tag{2.7}$$

Indeed,

$$P = D_\alpha^{-1/2} S D_\alpha^{1/2}.$$

The matrices P and S therefore have the same eigenvalues. Since S is symmetric, its eigenvectors are orthogonal and its eigendecomposition is numerically more stable than directly diagonalising P. If

$$S u_k = \lambda_k u_k,$$

then the corresponding right eigenvector of P is

$$\phi_k = D_\alpha^{-1/2} u_k.$$

The eigenvalues are ordered as

$$1 = \lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \cdots,$$

where the first eigenvector ϕ_0 is constant and carries no geometric information. The diffusion-map embedding at diffusion time t is therefore defined using the leading non-trivial eigenpairs:

$$\Psi_t(z_i) = \left(\lambda_1^t \phi_1(i), \ldots, \lambda_r^t \phi_r(i)\right) \in \mathbb{R}^r. \tag{2.8}$$

The above steps are presented in Algorithm 1.

Algorithm 1 Diffusion map

Input: $\tilde{Z} \in \mathbb{R}^{N \times D}$, ε, α, r
Output: Embedding coordinates $\Psi \in \mathbb{R}^{N \times r}$
$W_{ij} \leftarrow \exp(-\frac{\|z_i - z_j\|^2}{\varepsilon})$
$q_i \leftarrow \sum_{j=1}^N W_{ij}$
$W_{ij}^{(\alpha)} \leftarrow \frac{W_{ij}}{q_i^\alpha q_j^\alpha}$
$d_i^{(\alpha)} \leftarrow \sum_{j=1}^N W_{ij}^{(\alpha)}$
$D_\alpha \leftarrow \operatorname{diag}\left(d_1^{(\alpha)}, \ldots, d_N^{(\alpha)}\right)$
$P \leftarrow D_\alpha^{-1} W^{(\alpha)}$
$S \leftarrow D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2}$
Compute the eigendecomposition $S u_k = \lambda_k u_k$.
$\phi_k \leftarrow D_\alpha^{-1/2} u_k$
Sort the eigenpairs in descending order of λ_k.
Discard the trivial eigenpair (λ_0, ϕ_0).
Form $\Psi \leftarrow [\lambda_1 \phi_1, \ldots, \lambda_r \phi_r]$.
return Ψ

The eigenvalue factors suppress rapidly decaying modes, while the eigenvectors describe the dominant directions of diffusion through the data. With the complete eigendecomposition, Euclidean distance in diffusion-coordinate space is equal to diffusion distance:

$$D_t^2(z_i, z_j) = \sum_{k \geq 1} \lambda_k^{2t} \left(\phi_k(i) - \phi_k(j)\right)^2.$$

Truncating this expansion after r terms provides a low-dimensional approximation that retains the most persistent diffusion modes.

Under suitable assumptions on the sampling process, together with an appropriate relationship between the sample size and the bandwidth, the graph-based operator converges to a differential operator on $\mathcal{M}$. Correspondingly, its eigenvectors converge to evaluations of the eigenfunctions of the limiting operator at the sampled observations. This provides the theoretical basis for interpreting diffusion coordinates as empirical approximations to intrinsic spectral coordinates of the underlying system. In the latent-dynamical setting considered in Section 2.2, this connection motivates the use of diffusion maps to obtain data-driven coordinates for an otherwise unobserved state space.

The bandwidth ε is central to the construction. If ε is too small, the off-diagonal affinities become negligible, so that

$$W \approx I$$

and many observations are effectively isolated. In this regime, the diffusion operator describes a poorly connected collection of local components. If ε is too large, most affinities approach one, so that

$$W \approx \mathbf{1}\mathbf{1}^\top,$$

and the kernel loses its ability to distinguish local geometric structure. An appropriate bandwidth must therefore be sufficiently large to connect the observations while remaining sufficiently local to preserve the geometry of the data.

One diagnostic for the bandwidth is the kernel mass

$$T(\varepsilon) = \sum_{i,j=1}^{N} \exp\left(-\frac{\|z_i - z_j\|^2}{\varepsilon}\right). \tag{2.9}$$

For observations sampled from a sufficiently regular d-dimensional manifold, the kernel mass is expected to satisfy

$$T(\varepsilon) \approx C\varepsilon^{d/2}$$

over an appropriate range of local scales. It follows that

$$\frac{d \log T(\varepsilon)}{d \log \varepsilon} \approx \frac{d}{2}. \tag{2.10}$$

A stable region in the log–log slope can therefore provide information about both the intrinsic dimension and a suitable range of bandwidths. In finite, noisy, and non-uniformly sampled data, however, such a scaling region may be weak or absent. Kernel-mass scaling is therefore best treated as a diagnostic and considered alongside graph connectivity, the degree distribution, spectral stability and the quality of the resulting coordinates.

The embedding dimension r determines how many non-trivial diffusion modes are retained. A common diagnostic is the presence of a spectral gap. If

$$\lambda_r - \lambda_{r+1}$$

is large relative to neighbouring gaps, the first r non-trivial eigenvectors represent comparatively persistent modes, whereas subsequent eigenvectors decay more rapidly and may primarily reflect fine-scale variation or noise. Spectral gaps do not provide an infallible choice of dimension, particularly when eigenvalues occur in near-degenerate groups. The final dimension should therefore also be assessed through stability under nearby bandwidths, preservation of relevant geometric structure, and the requirements of the downstream analysis.

2.4 Kernel scale and spectral diagnostics

Explanation of the theory behind choosing our model parameters basically. The kernel mass plots, spectral analysis, and the justification of the embedding dimension. Also graph connectivity and robustness of all choices.

- kernel mass $T(\varepsilon)$;
- the scaling relation $T(\varepsilon) \propto \varepsilon^{d/2}$;
- spectral gaps;
- eigenvalue multiplicities;
- graph connectivity;
- stability across nearby parameter values.

2.5 Graph geometry and shortest paths

Explain how we construct our graph on the diffusion coordinates.

- Graph paths;
- ordinary graph-geodesic cost;
- the distinction between a Euclidean chord and a path constrained to observed neighbourhoods;
- Dijkstra's shortest-path problem.

2.6 Density-aware graph geometry??

This is where you can talk about the link between density and plausibility. Go into extra detail here because this is novel

- regions visited frequently by history correspond to high density
- low density means combinations that have not been seen before.
- show the negative correlation with distance and density.

2.7 Hidden Markov models??

This is the extension, and I am not sure if it will be done

3 Data and Preprocessing

3.1 Data sources and sample period

FRED-MD, Kenneth-French Library, NBER recession indicators

3.2 Variable selection

3.3 Temporal alignment and missing observations

3.4 Transformations

3.5 Data diagnostics

3.6 Final analysis matrix (and sample splits, if HMM)??

4 Numerical methods

4.1 Overview of the computational pipeline

This is where you give a clear picture of what you actually are going to do. Reference the different sections, etc.

raw data $\to$ transformation $\to$ diffusion map $\to$ density and graph $\to$ stress paths $\to$ lifting

? $\to$ HMM regime analysis

In this section, we describe the entire computational pipeline. Firstly, using our transformed and standardised data, we find a diffusion map, which captures the potential non-linear structure of the macroeconomic manifold on which the data lies. Using that diffusion map, we choose an appropriate embedding dimension, and now we have our new coordinate system. Using these new coordinates, we construct a k-nearest-neighbours graph and compute an estimation of the densities. We then implement our density-aware graph path model to interpolate a path between a variety of specifically chosen economic endpoints. We compare this path to various other types of paths, specifically, a linear path in diffusion space (Is this also a linear path in macro space?), and a standard graph path. Then we can lift these paths back to the macro space, and there we can compare different paths in real coordinates to see how well our model does.

4.2 Diffusion-map implementation - Example

The diffusion-map construction described in Section 2.3 was applied to the transformed macroeconomic state matrix

$$\widetilde{Z} = \begin{pmatrix} \widetilde{z}_1^\top \\ \vdots \\ \widetilde{z}_N^\top \end{pmatrix} \in \mathbb{R}^{N\times D},$$

where each row represents one monthly observation and each column corresponds to a macroeconomic or financial variable. The transformation and construction of this matrix are described in Section 3. The NBER recession indicator was retained for subsequent validation but was not included as an input to the embedding.

Pairwise squared Euclidean distances were first calculated between all transformed observations. These distances were converted into Gaussian affinities using the diffusion bandwidth ε. The density-normalisation parameter was fixed at $\alpha = 1$. This choice reduces the influence of variations in the empirical sampling density, so that the resulting operator is intended to reflect the geometry of the underlying state space rather than simply assigning greater importance to densely sampled periods. We now have the matrix $W^{(\alpha)}$ from which we construct the diagonal degree matrix D_α and the transition matrix $P = D_\alpha^{-1}W^{(\alpha)}$ following the steps from Section 2.3.

For numerical stability, the eigenpairs were obtained from the symmetric matrix

$$S = D_\alpha^{-1/2} W^{(\alpha)} D_\alpha^{-1/2},$$

rather than by directly diagonalising the generally non-symmetric transition matrix P. The eigenvalues were arranged in descending order and the corresponding right eigenvectors of the transition matrix were recovered from the symmetric eigenvectors. The first eigenvector, associated with the trivial eigenvalue $\lambda_0 = 1$, was omitted. At diffusion time $t = 1$, the r-dimensional coordinate assigned to observation i was

$$\Psi(\widetilde{z}_i) = \big(\lambda_1\phi_1(i), \dots, \lambda_r\phi_r(i)\big).$$

Thus, the embedding retained both the leading non-trivial eigenvectors and their associated eigenvalue scaling. The above was a brief outline of Algorithm 1.

4.3 Diffusion parameter selection

The kernel bandwidth ε controls the locality of the Gaussian affinity kernel and was selected before fixing the embedding dimension. As an initial scale diagnostic, the kernel mass

$$T(\varepsilon) = \sum_{i,j} \exp\left(-\frac{\|\tilde{z}_i - \tilde{z}_j\|^2}{\varepsilon}\right)$$

was evaluated over a logarithmically spaced range of bandwidths, together with its local log–log slope, as described in Section 2.4. This diagnostic identifies scales at which the total kernel mass changes most rapidly. However, in a finite and non-uniformly sampled dataset, the bandwidth selected from the slope alone may produce a poorly connected graph. The kernel-mass rule was therefore used to identify a plausible bandwidth range rather than as an automatic selection criterion.

Candidate bandwidths were subsequently compared using the connectivity and spectral properties of the resulting diffusion operator. Let

$$q_i(\varepsilon) = \sum_j W_{ij}(\varepsilon)$$

denote the unnormalised kernel degree of observation i. The minimum degree,

$$q_{\min}(\varepsilon) = \min_i q_i(\varepsilon),$$

was used to detect observations that were effectively isolated from the remainder of the sample. Since $W_{ii} = 1$, a value of q_i close to one indicates that almost all of the degree is contributed by the self-affinity. The heterogeneity of the degree distribution was measured using

$$\mathrm{CV}_q(\varepsilon) = \frac{\mathrm{sd}\{q_i(\varepsilon)\}}{\mathrm{mean}\{q_i(\varepsilon)\}}.$$

A large coefficient of variation indicates that the graph connectivity is concentrated unevenly across observations, while comparison across neighbouring bandwidths provides a check on the stability of the connectivity structure.

The leading non-trivial eigenvalues of the transition matrix P were also compared across candidate bandwidths. Particular attention was given to the stability of the dominant eigenvalues and the gaps between consecutive eigenvalues under small changes in ε. The corresponding diffusion coordinates were inspected to exclude bandwidths for which the leading eigenvectors were nearly constant over most observations but contained isolated sharp spikes, as this behaviour is indicative of weakly connected or nearly isolated graph components.

Combining these diagnostics, the bandwidth was fixed at

$$\varepsilon = 3.$$

This choice provided adequate graph connectivity while retaining sufficient locality in the kernel. The kernel-mass diagnostic, degree statistics and spectral comparisons supporting this choice are reported in Section 6.1.

The embedding dimension was not selected from the eigenspectrum alone. Candidate dimensions were assessed using three complementary criteria: the leading non-trivial eigenspectrum, the separation of near-collision pairs and the reconstruction performance of the local-neighbour lifting procedure, outlined in Section 4.7.

The eigenspectrum was used to identify the number of dominant diffusion modes and to locate large gaps between consecutive non-trivial eigenvalues. Since a spectral gap does not by itself guarantee that the retained coordinates distinguish all economically different observations, an additional near-collision diagnostic was introduced. For an embedding of dimension r, pairs of observations separated by no fewer than twelve months were ranked according to their Euclidean distance in diffusion space. The closest 1% of eligible pairs were defined as near-collisions. Their distance ranks were then recomputed after adding coordinate $r + 1$. The proportion of these pairs that left the closest 1%, 5% or 10% in the higher-dimensional embedding measured how frequently the additional coordinate separated observations that had appeared nearly identical in the lower-dimensional representation. A large exit proportion therefore indicates that coordinate $r + 1$ contains substantial information not represented by the first r coordinates.

The ability of each candidate embedding to reconstruct the transformed macroeconomic observations was assessed through the local-neighbourhood lift. If $\widehat{Z}^{(r)}$ denotes the observations reconstructed from an r-dimensional embedding, reconstruction accuracy was summarised using the root mean squared error and the proportion of variance recovered,

$$R^2_{\text{lift}}(r) = 1 - \frac{\|Z - \widehat{Z}^{(r)}\|_F^2}{\|Z - \overline{Z}\|_F^2},$$

where $\overline{Z}$ contains the variable-wise sample means. This diagnostic measures the extent to which the retained diffusion coordinates preserve information required to recover the original transformed variables.

Finally, low-dimensional projections were inspected after colouring observations by calendar time, recession status and selected macroeconomic variables. These visualisations were used as a qualitative check that the retained coordinates represented distinct and economically interpretable directions of variation, rather than as a standalone dimension-selection rule.

The combined diagnostics led to the selection

$$r = 4.$$

The fourth coordinate provided a material improvement in near-collision separation and lifted reconstruction relative to the three-dimensional embedding, whereas the incremental benefit of adding a fifth coordinate was comparatively small. The numerical comparisons and embedding visualisations are presented in Section 6.1.

4.4 Graph construction and density estimation

Following the construction of the diffusion embedding, an undirected k-nearest-neighbours graph was formed from the observed diffusion coordinates,

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \qquad \mathcal{V} = \{1, \dots, N\}.$$

A directed edge from node i to node j was first introduced whenever ψ_j was among the k nearest neighbours of ψ_i in diffusion space. The graph was then symmetrised by taking the union of the directed edge sets, so that $(i, j) \in \mathcal{E}$ whenever either observation selected the other as a neighbour. Consequently, the average degree of the resulting graph may be slightly greater than k. The Euclidean length assigned to each edge was

$$\ell_{ij} = \|\psi_i - \psi_j\|.$$

The nearest-neighbour search was implemented using Scikit-Learn's `NearestNeighbors` class.

Candidate values of k were assessed using several complementary graph diagnostics. First, the graph was required to be connected, ensuring that a path existed between any selected pair of endpoints. All candidate values considered satisfied this condition. Second, the graph density

$$d_{\mathcal{G}} = \frac{2|\mathcal{E}|}{N(N-1)}$$

was monitored to ensure that the graph remained local and did not approach a complete graph as k increased.

The presence of unusually long edges within shortest paths was also examined. A systematic subsample containing every fifth node was formed, and shortest paths were calculated between every unordered pair of nodes in this subsample. For each path Γ, the statistic

$$R_{\text{edge}}(\Gamma) = \frac{\max_{(i,j)\in\Gamma} \ell_{ij}}{\text{median}_{(i,j)\in\Gamma} \ell_{ij}}$$

was recorded. A large value indicates that the path depends upon an edge whose length is substantially greater than its typical edge length. Such an edge may act as an artificial shortcut across a sparsely sampled region, causing the graph path to represent a non-local jump rather than gradual movement through the sampled manifold. The distribution of this statistic was therefore compared across candidate values of k.

Finally, the resulting paths and their principal conclusions were required to remain stable under nearby choices of k. These diagnostics were used to select the baseline graph specification, while a broader sensitivity analysis with respect to k is reported in the robustness analysis.

A relative density score was then estimated from the observations in diffusion space. For an arbitrary point $\psi \in \mathbb{R}^r$, define

$$\widehat{\rho}(\psi; h) = \sum_{j=1}^{N} \exp\left(-\frac{\|\psi - \psi_j\|^2}{h}\right),$$

where $h > 0$ is another bandwidth parameter which controls the locality of the density estimate. At an observed node, this gives

$$\widehat{\rho}_i = \widehat{\rho}(\psi_i; h).$$

Since only relative density is required for the path construction, the kernel score was normalised by its mean over the observed nodes,

$$\bar{\rho} = \frac{1}{N}\sum_{i=1}^{N} \widehat{\rho}_i, \qquad \rho(\psi) = \frac{\widehat{\rho}(\psi; h)}{\bar{\rho}},$$

with $\rho_i = \rho(\psi_i)$.

Candidate values of h were compared according to whether the resulting density field distinguished well-supported and sparsely supported observations without becoming nearly constant or being dominated by isolated spikes. Stability under nearby bandwidth choices was also examined. The selected value and its sensitivity are reported with the empirical results and robustness checks.

For numerical stability, a small density floor $\delta > 0$ was imposed,

$$\rho_{i,\delta} = \max\{\rho_i, \delta\},$$

and the density potential at node i was defined by

$$V_i = -\log(\rho_{i,\delta}).$$

Under this construction, observations surrounded by many nearby historical states have relatively low potential, whereas isolated observations have relatively high potential.

The density score is interpreted as a measure of *historical support under the geometry learned by the diffusion map*. It does not by itself establish that a state is economically plausible or dynamically attainable. Its empirical relationship with recognised periods of economic stress and with proximity to historical observations in the transformed macroeconomic space is examined in the results.

I think I can be less conservative here; there is a link between real support and density. Depends on what I put in the background, but I think the above may be underselling it slightly

4.5 Stress-path construction

4.5.1 Endpoint selection

Four historically motivated start–stress endpoint pairs were considered. Both the start and stress endpoints were required to correspond to observed dates and hence to existing nodes of the diffusion graph. No out-of-sample extension was therefore required for the principal stress-path analysis.

The endpoints were selected to represent economically interpretable transitions from comparatively benign macroeconomic conditions to distinct forms of historical stress. The specific dates, scenario labels and summary economic characteristics are introduced in the empirical results. More detailed historical discussion is provided separately where required.

4.5.2 Path definitions

Three paths were constructed between each pair of endpoints: a linear path through diffusion space, an unpenalised graph shortest path and a density-aware graph shortest path. Comparing these constructions separates the effect of restricting movement to the empirical graph from the additional effect of favouring historically well-supported regions.

Let i_{start} and i_{stress} denote the graph nodes corresponding to the selected endpoints. A graph path is represented by an ordered sequence of node indices

$$\Gamma = (i_1, \ldots, i_L),$$

where

$$i_1 = i_{\text{start}}, \qquad i_L = i_{\text{stress}}, \qquad (i_m, i_{m+1}) \in \mathcal{E}.$$

For a specified set of positive edge costs w_{ij}, the shortest path solves

$$\Gamma^\star = \underset{\Gamma: i_{\text{start}} \to i_{\text{stress}}}{\arg\min} \sum_{m=1}^{L-1} w_{i_m i_{m+1}}.$$

The optimisation was implemented using Dijkstra's Algorithm via SciPy's graph shortest-path routine.

The unpenalised graph path assigns each edge its Euclidean diffusion-space length,

$$w_{ij}^{(0)} = \ell_{ij} = \|\psi_i - \psi_j\|.$$

This path is constrained to move through observed graph nodes but does not explicitly distinguish between densely and sparsely supported regions.

To incorporate historical support, the density-aware edge cost was defined for $\beta \geq 0$ by

$$w_{ij}^{(\beta)} = \|\psi_i - \psi_j\| \exp\left(\beta \frac{V_i + V_j}{2}\right).$$

The average of the endpoint potentials approximates the density potential encountered while traversing the edge. Since low-density nodes have larger values of V_i, edges passing through sparsely supported regions receive a higher cost. The corresponding density-aware path is

$$\Gamma_\beta = \underset{\Gamma: i_{\text{start}} \to i_{\text{stress}}}{\arg\min} \sum_{(i,j) \in \Gamma} w_{ij}^{(\beta)}.$$

When $\beta = 0$, this reduces to the unpenalised graph shortest path. Increasing β places progressively greater emphasis on historical support relative to direct path length.

Candidate values of β were compared according to the trade-off between improved density support and the additional length or detour introduced into the path. Stability across nearby values was also examined. The retained specification is used for the principal analysis, while the sensitivity of the results to β is considered in the robustness analysis.

As an unconstrained benchmark, a straight line was constructed between the same endpoints in diffusion space,

$$\gamma_{\text{lin}}(s) = (1-s)\psi_{\text{start}} + s\psi_{\text{stress}}, \qquad s \in [0,1].$$

The path was evaluated on an equally spaced grid of values of s. Unlike the graph paths, its intermediate points need not coincide with observed historical states and it may pass through regions containing little empirical support. It therefore provides a diffusion coordinate space linear benchmark against which the effects of graph restriction and density weighting can be assessed.

The graph and linear paths are compared using the path-length, density-support and macroeconomic evaluation measures defined in Section 4.6.

4.6 Stress-path evaluation

The stress paths are evaluated according to three complementary properties: the historical support encountered along the path, the distance travelled through diffusion space, and the regularity of the resulting trajectory. These quantities are considered jointly, since a path may avoid sparsely supported regions only by taking a substantially longer or more irregular route.

For evaluation, let a discretised path be written as

$$\gamma = (\gamma_0, \dots, \gamma_L),$$

where $\gamma_\ell \in \mathbb{R}^r$. For a graph path, the points γ_ℓ are observed diffusion coordinates ψ_{i_ℓ}, whereas for the linear benchmark they are interpolated points along the straight line defined in Section 4.5.

The principal evaluation criterion is the density encountered along the path. For an interpolated point γ_ℓ that does not coincide with an observed state, the density score is evaluated against the historical diffusion coordinates using the same kernel and bandwidth as in Section 4.4,

$$\widehat{\rho}(\gamma_\ell) = \sum_{j=1}^{N} \exp\left(-\frac{\|\gamma_\ell - \psi_j\|^2}{h}\right).$$

This is normalised using the same full-sample scale

$$\overline{\rho} = \frac{1}{N}\sum_{i=1}^{N} \widehat{\rho}(\psi_i),$$

giving

$$\rho(\gamma_\ell) = \frac{\widehat{\rho}(\gamma_\ell)}{\overline{\rho}}.$$

A leave-one-out modification is used whenever the evaluated point corresponds to an observed graph node. For $\gamma_\ell = \psi_i$, its evaluation density is

$$\widehat{\rho}_{-i}(\psi_i) = \sum_{\substack{j=1 \\ j\neq i}}^{N} \exp\left(-\frac{\|\psi_i - \psi_j\|^2}{h}\right), \qquad \rho_{-i}(\psi_i) = \frac{\widehat{\rho}_{-i}(\psi_i)}{\overline{\rho}}.$$

Removing the observation's self-contribution prevents graph nodes from receiving an automatic affinity of one purely because they belong to the historical sample. This leave-one-out adjustment is used only for path evaluation; the density field used to construct the density-aware graph remains that defined in Section 4.4. The same convention is applied to any observed endpoint appearing on the linear path. After evaluation, the density floor δ is applied as before.

Denote the resulting evaluation density at γ_ℓ by $\rho_{\text{eval}}(\gamma_\ell)$. The primary support metric is the minimum density,

$$M_{\min}(\gamma) = \min_{0 \le \ell \le L} \rho_{\mathrm{eval}}(\gamma_\ell),$$

which identifies the least historically supported state encountered along the path. Since this metric is determined by a single point, the overall level of support is also summarised by the path-average density. To account for differences in path discretisation, this is weighted by the diffusion-space length of each segment,

$$M_{\mathrm{mean}}(\gamma) = \frac{1}{L_{\mathrm{diff}}(\gamma)} \sum_{\ell=0}^{L-1} \frac{\rho_{\mathrm{eval}}(\gamma_\ell) + \rho_{\mathrm{eval}}(\gamma_{\ell+1})}{2} \|\gamma_{\ell+1} - \gamma_\ell\|,$$

where

$$L_{\mathrm{diff}}(\gamma) = \sum_{\ell=0}^{L-1} \|\gamma_{\ell+1} - \gamma_\ell\|$$

is the total path length in diffusion coordinate space. This provides a measure of the additional distance required for a path to remain in better-supported regions of the learned diffusion geometry.

Finally, path regularity is measured through changes in direction between successive segments. Define the unit direction vector

$$u_\ell = \frac{\gamma_{\ell+1} - \gamma_\ell}{\|\gamma_{\ell+1} - \gamma_\ell\|}, \qquad \ell = 0, \dots, L-1.$$

The turning angle at an interior path point is

$$\theta_\ell = \arccos\left(u_{\ell-1}^\top u_\ell\right), \qquad \ell = 1, \dots, L-1,$$

and the total direction change is

$$T_{\mathrm{turn}}(\gamma) = \sum_{\ell=1}^{L-1} \theta_\ell.$$

A straight path has zero direction change, while larger values indicate a more tortuous trajectory through diffusion space.

The path constructions are therefore compared using minimum and average historical support together with path length in diffusion coordinate space and mean direction change. The objective is not to optimise any one of these quantities independently, but to assess whether the graph-based constructions achieve greater historical support without requiring excessive detours or irregularity. The empirical comparisons are reported in Section 6.3.

4.7 Lifting paths into macroeconomic space

The stress paths constructed in diffusion space must ultimately be expressed in terms of the transformed macroeconomic variables. For graph paths, no estimated inverse mapping is required. Each graph vertex corresponds to an observed diffusion coordinate ψ_i and therefore to its associated macroeconomic observation $\tilde{z}_i$. Hence, for a graph path

$$\Gamma = (i_1, \dots, i_L),$$

the corresponding macroeconomic path is obtained directly as

$$(\tilde{z}_{i_1}, \dots, \tilde{z}_{i_L}).$$

In contrast, the intermediate points of the linear benchmark generally do not correspond to observed states and must therefore be lifted from diffusion space back to macroeconomic space.

Two lifting procedures were considered. The first is a global linear inverse map, following the approach of Baker et al. [1]. Let

$$\psi_i^{(c)} = \psi_i - \overline{\psi}, \qquad \widetilde{z}_i^{(c)} = \widetilde{z}_i - \overline{z},$$

denote the centred diffusion and macroeconomic coordinates. The lifting matrix is obtained from

$$H = \arg\min_{B \in \mathbb{R}^{D \times r}} \sum_{i=1}^{N} \left\| \widetilde{z}_i^{(c)} - B\psi_i^{(c)} \right\|^2,$$

and an arbitrary diffusion-space point ψ is lifted according to

$$\widehat{z}_{\text{lin}}(\psi) = H(\psi - \overline{\psi}) + \overline{z}.$$

Centring the variables before fitting and subsequently restoring $\overline{z}$ is equivalent to allowing the linear reconstruction to contain an intercept. This construction will be referred to as the Baker-style lift.

The second procedure is a local-neighbourhood lift (LNL). For a point ψ, let $\mathcal{N}_m(\psi)$ denote its m nearest historical observations in diffusion space. The lifted state is defined as the weighted average

$$\widehat{z}_{\text{LNL}}(\psi) = \sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi)\widetilde{z}_j,$$

where

$$a_j(\psi) = \frac{\exp\left(-\|\psi - \psi_j\|^2/\tau\right)}{\sum\limits_{k \in \mathcal{N}_m(\psi)} \exp\left(-\|\psi - \psi_k\|^2/\tau\right)}.$$

The weights are non-negative and satisfy

$$\sum_{j \in \mathcal{N}_m(\psi)} a_j(\psi) = 1.$$

Unlike the global linear lift, this construction allows the inverse approximation to vary across different regions of the embedding.

The scale parameter τ was fixed using the median squared distance between neighbouring observations in diffusion space, providing a characteristic local distance scale for the Gaussian weights. The neighbourhood size m was selected by comparing leave-one-out reconstruction error across a range of candidate values. The resulting parameter choice and reconstruction results are reported in Section 6.4.

Since the endpoints of each linear stress path correspond to known historical observations, their macroeconomic values are available directly. The lifted linear paths were therefore anchored to the exact observed start and stress states rather than using the reconstructed values produced by either lifting method at these two points.

To assess the ability of each lifting procedure to recover historical observations, leave-one-out reconstruction diagnostics were calculated. For each observation i, the corresponding macroeconomic state was removed from the training sample and the lifting procedure was constructed using the remaining $N-1$ observations. The held-out diffusion coordinate ψ_i was then lifted to obtain $\widehat{z}_{-i}$ and compared with the true state $\widetilde{z}_i$. Reconstruction performance was summarised using the root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{ND} \sum_{i=1}^{N} \|\widetilde{z}_i - \widehat{z}_{-i}\|^2},$$

and the proportion of variance recovered,

$$R^2_{\text{lift}} = 1 - \frac{\sum_{i=1}^{N} \|\widetilde{z}_i - \widehat{z}_{-i}\|^2}{\sum_{i=1}^{N} \|\widetilde{z}_i - \overline{z}\|^2}.$$

These quantities were compared with a baseline reconstruction that assigns each held-out observation the mean of the remaining training observations. As the leave-one-out error is also used in selecting the neighbourhood size for LNL, these quantities are treated as cross-validated reconstruction diagnostics rather than as an entirely independent test set.

A second diagnostic was used to detect whether either lifting procedure mapped the interpolated linear paths unusually far from the historical macroeconomic cloud. For an interior lifted path point $\widehat{z}_\ell$, define

$$d_{\text{NN}}(\widehat{z}_\ell) = \min_{1 \le j \le N} \|\widehat{z}_\ell - \widetilde{z}_j\|.$$

The median and 95th percentile of these distances were calculated along each lifted linear path. The fixed historical endpoints were excluded from this diagnostic, since their nearest-neighbour distance is zero by construction.

To provide a reference scale, these path distances were compared with leave-one-out nearest-neighbour distances among the historical observations,

$$d_{\text{NN},-i}(\widetilde{z}_i) = \min_{j \ne i} \|\widetilde{z}_i - \widetilde{z}_j\|.$$

The corresponding historical median and 95th percentile indicate the typical and upper-tail separation between observed macroeconomic states. The lifted-path diagnostic therefore tests whether the inverse mapping produces states lying unusually far from the empirical macroeconomic cloud. It is not interpreted as a direct measure of economic plausibility.

The lifting procedure used in the principal stress-path analysis was selected by considering both cross-validated reconstruction accuracy and the extent of extrapolation observed along the lifted linear paths. The numerical comparison between the Baker-style and local-neighbourhood lifts is presented in Section 6.4.

4.8 Nyström extension for out-of-sample embedding

The Nyström extension provides a way to embed a new macroeconomic observation into the previously learned diffusion space without recomputing the diffusion map or its eigendecomposition. The original embedding is therefore kept fixed, and the location of the new point is determined from its affinities to the historical observations.

Suppose that a new transformed macroeconomic observation

$$\widetilde{z}_{\text{new}} \in \mathbb{R}^D$$

is to be embedded into the r-dimensional diffusion space. The extension uses the bandwidth ε, training degrees q_i, and diffusion-map eigenpairs $\{(\lambda_\ell, \phi_\ell)\}$ obtained from the original dataset $\widetilde{Z}$. Here, ϕ_ℓ denotes the right eigenvector of the transition operator associated with λ_ℓ.

First, the affinity between the new point and each historical observation is calculated using the same Gaussian kernel as in the original diffusion map,

$$w_{\text{new},i} = \exp\left(-\frac{\|\widetilde{z}_{\text{new}} - \widetilde{z}_i\|^2}{\varepsilon}\right), \qquad i = 1, \ldots, N.$$

The corresponding kernel degree of the new observation is

$$q_{\text{new}} = \sum_{i=1}^{N} w_{\text{new},i}.$$

To remain consistent with the diffusion-map construction, the same α-normalisation is then applied,

$$w^{(\alpha)}_{\text{new},i} = \frac{w_{\text{new},i}}{q^{\alpha}_{\text{new}} q^{\alpha}_{i}}.$$

These affinities are row-normalised to produce a transition probability from the new point to each historical observation,

$$p_{\text{new},i} = \frac{w^{(\alpha)}_{\text{new},i}}{\sum_{j=1}^{N} w^{(\alpha)}_{\text{new},j}}.$$

The intuition behind the Nyström extension follows directly from the eigenvector equation for the transition matrix. For a historical observation i,

$$\lambda_\ell \phi_\ell(i) = \sum_{j=1}^{N} P_{ij}\phi_\ell(j).$$

Thus, the value of the ℓth eigenfunction at a point is related to a weighted average of its values at neighbouring observations. For the new point, the row $p_{\text{new},i}$ plays the role of a new transition row. Requiring the same eigenvector relation to hold gives

$$\lambda_\ell \phi_\ell(\widetilde{z}_{\text{new}}) = \sum_{i=1}^{N} p_{\text{new},i}\phi_\ell(i),$$

and hence

$$\phi_\ell(\widetilde{z}_{\text{new}}) = \frac{1}{\lambda_\ell}\sum_{i=1}^{N} p_{\text{new},i}\phi_\ell(i).$$

The new observation therefore inherits its diffusion coordinates from the eigenfunction values of nearby historical observations, weighted according to its position relative to the learned data geometry. Importantly, the new point does not alter the existing eigenvectors or diffusion coordinates.

Using the same diffusion-time convention as in the original embedding, the out-of-sample diffusion coordinates are

$$\Psi(\widetilde{z}_{\text{new}}) = \left(\lambda_1\phi_1(\widetilde{z}_{\text{new}}), \ldots, \lambda_r\phi_r(\widetilde{z}_{\text{new}})\right).$$

This procedure follows the Nyström extension principle used in [3], with the additional α-normalisation required to remain consistent with the diffusion-map construction used throughout this dissertation.

4.9 Gaussian HMM specification and evaluation

The diffusion-map analysis describes the geometry and historical support of macroeconomic states, but does not explicitly model their temporal evolution. As an additional analysis, a Gaussian Hidden Markov Model (HMM) was therefore fitted to the sequence of diffusion coordinates. The HMM is used to investigate whether observations in the learned embedding can be organised into persistent economic regimes and, subsequently, whether the constructed stress paths are consistent with the temporal structure learned from the historical data.

Let $S_t \in \{1, \ldots, K\}$ denote the latent state at time t. Conditional on $S_t = k$, the diffusion coordinates are assumed to follow a multivariate Gaussian distribution,

$$\Psi_t \mid S_t = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

where a full covariance matrix Σ_k is estimated separately for each state. The latent states evolve according to the first-order Markov transition probabilities

$$A_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i).$$

The general HMM construction and sequence likelihood are described in the mathematical background.

The number of hidden states was selected by comparing models with

$$K \in \{2, 3, 4, 5, 6, 7\}.$$

For each K, the model was fitted from 100 different random initialisations in order to reduce sensitivity to local optima in the likelihood surface, and the fit attaining the largest log-likelihood was retained. For an r-dimensional diffusion embedding, the number of free parameters in the full-covariance Gaussian HMM is

$$p_K = (K-1) + K(K-1) + Kr + K\frac{r(r+1)}{2}.$$

These terms correspond respectively to the initial-state distribution, transition matrix, state-dependent means and state-dependent covariance matrices.

Model complexity was first assessed using the Bayesian Information Criterion,

$$\mathrm{BIC}_K = p_K \log N - 2 \log L_K,$$

where $\log L_K$ is the maximised log-likelihood of the observed diffusion-coordinate sequence and N is the number of observations. Lower BIC values are preferred, since an increase in model complexity must provide a sufficient improvement in likelihood to offset the additional parameters.

Predictive fit was assessed separately using a chronological training–test split. The diffusion map was fitted using the first 70% of the observations, producing the training coordinates Ψ_{train}. The remaining 30% were embedded into this fixed diffusion space using the Nyström extension described in Section 4.8, giving Ψ_{test}. For each candidate K, an HMM was fitted to Ψ_{train} and the log-likelihood assigned to the held-out sequence Ψ_{test} was recorded. The BIC and out-of-sample likelihood need not favour the same model, and the resulting model-selection comparison is therefore discussed in the HMM results rather than imposing either criterion mechanically.

After selecting a specification, the inferred states were characterised using the corresponding macroeconomic observations. The most likely hidden-state sequence,

$$\widehat{S}_{1:N},$$

was obtained from the fitted HMM, and the macroeconomic variables associated with each state were summarised to determine whether the statistical regimes admitted meaningful economic interpretations.

The inferred states were also compared with the NBER recession chronology. For each state k, define the binary indicator

$$I_t^{(k)} = \mathbb{1}\{\widehat{S}_t = k\},$$

and let $R_t \in \{0, 1\}$ denote the NBER recession indicator. Agreement between $I_t^{(k)}$ and R_t was measured using Cohen's κ,

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is the observed proportion of agreement and p_e is the agreement expected from the marginal frequencies of the two binary series. Unlike raw classification accuracy, κ therefore accounts for agreement that would arise from the relative frequency of recession and non-recession observations.

Once a state exhibiting the clearest crisis or recession interpretation had been identified from the economic characteristics and external comparisons, its posterior probability was used for an additional lead–lag diagnostic. Let

$$\pi_t^{(c)} = \mathbb{P}\left(S_t = c \mid \Psi_{1:N}\right)$$

denote the posterior probability of the selected crisis state. For offsets

$$\ell \in \{-6, \ldots, 6\},$$

the correlation

$$C(\ell) = \mathrm{Corr}\left(\pi_t^{(c)}, R_{t+\ell}\right)$$

was calculated over the common time indices. Under this convention, $\ell > 0$ tests whether elevated crisis-state probabilities precede the corresponding NBER indicator, whereas $\ell < 0$ tests whether the HMM state tends to lag the NBER chronology. Since the posterior probabilities are conditioned on the complete observed sequence, this analysis is interpreted as a descriptive lead–lag diagnostic rather than an out-of-sample forecasting test.

The relationship between the HMM states and the empirical support of the diffusion embedding was also examined by grouping the diffusion-space density, or equivalently the density potential

$$V_t = -\log \rho_t,$$

according to the inferred state. This determines whether the statistical regimes occupy systematically different regions of the empirical density distribution. The economic interpretation of low-density observations is established separately in Section 6.2; consequently, the state-wise density comparison is treated as a coherence diagnostic rather than as an independent validation criterion.

Finally, we link this section back to the main theme of this dissertation by using our fitted HMM to provide a temporal consistency score for the stress paths constructed in Section 4.5. For a discretised path

$$\gamma = (\gamma_0, \ldots, \gamma_L),$$

the fitted model assigns the sequence likelihood

$$p(\gamma_0, \ldots, \gamma_L \mid \widehat{\theta}),$$

where $\widehat{\theta}$ denotes the estimated HMM parameters. Since the linear, ordinary graph and density-aware graph paths may contain different numbers of vertices, the comparison is based on the log-likelihood per path point,

$$\mathcal{S}_{\mathrm{HMM}}(\gamma) = \frac{1}{L+1} \log p(\gamma_0, \ldots, \gamma_L \mid \widehat{\theta}).$$

Larger values indicate that the sequence is more compatible with the regime distributions and transitions learned from the historical diffusion-coordinate series.

The vertices of a stress path do not necessarily represent equal monthly increments, particularly for graph paths. The HMM score is therefore used as a relative measure of consistency with the learned temporal structure rather than as the literal probability that the economy follows the scenario at monthly frequency. The model-selection results, economic state interpretations, NBER comparisons, density diagnostics and stress-path scores are reported in the dedicated HMM results section.

4.10 Principal component benchmark

To provide a conventional linear dimensionality-reduction benchmark for the proposed manifold-based stress paths, principal component analysis (PCA) was applied to the same quantile-standardised macroeconomic observations used to construct the diffusion map. The first $r = 4$ principal components were retained, matching the dimension of the diffusion embedding and thereby allowing the comparison to focus on the distinction between a linear PCA representation and the nonlinear diffusion embedding rather than on the number of retained coordinates.

Let

$$\Phi_{\mathrm{PCA}} : \mathbb{R}^D \to \mathbb{R}^r$$

denote the fitted PCA transformation and let

$$y_i = \Phi_{\mathrm{PCA}}(z_i)$$

be the PCA coordinates of observation z_i. For each of the start–stress event pairs introduced in Section 4.5, a linear path was constructed between the corresponding PCA coordinates,

$$\gamma_{\mathrm{PCA}}(s) = (1 - s)y_{\mathrm{start}} + s y_{\mathrm{stress}}, \qquad s \in [0, 1].$$

The path was discretised using the same number of points as the corresponding density-aware graph path. This choice affects only the numerical representation of the continuous PCA interpolation and provides a comparable resolution for the subsequent diagnostics and plots.

Each interpolated point was mapped back to the quantile-standardised macroeconomic space using the inverse PCA transformation,

$$\widehat{z}_{\mathrm{PCA}}(s) = \Phi_{\mathrm{PCA}}^{-1}\big(\gamma_{\mathrm{PCA}}(s)\big).$$

Since only the first four principal components are retained, this inverse transformation reconstructs observations within the rank-4 PCA subspace and does not in general reproduce the original event endpoints exactly. The PCA endpoints were therefore left as their genuine rank-4 reconstructions rather than being manually replaced by the observed macroeconomic states. The resulting loss of information was quantified for each event using the mean endpoint reconstruction error

$$E_{\mathrm{end}} = \frac{1}{2}\left(\|\widehat{z}_{\mathrm{PCA}}(0) - z_{\mathrm{start}}\|_2 + \|\widehat{z}_{\mathrm{PCA}}(1) - z_{\mathrm{stress}}\|_2\right).$$

This provides a direct measure of how accurately the retained PCA representation captures the particular benign and stressed states used in the scenario analysis.

The location of the PCA path relative to the empirical macroeconomic data cloud was assessed using the nearest-neighbour diagnostic introduced in Section 4.7. For an interpolated PCA reconstruction $\widehat{z}_{\mathrm{PCA}}(s)$,

$$d_{\mathrm{NN}}\big(\widehat{z}_{\mathrm{PCA}}(s)\big) = \min_j \|\widehat{z}_{\mathrm{PCA}}(s) - z_j\|_2.$$

Distances were calculated in quantile-standardised macroeconomic space and compared with the leave-one-out nearest-neighbour distances of the historical observations. The median, 95th percentile and maximum path distances were recorded. As in the lifting analysis, this diagnostic is not interpreted as a measure of economic plausibility; rather, it identifies whether the generated path passes through regions unusually distant from the observed macroeconomic sample.

For economic interpretation, the PCA reconstructions were subsequently mapped to the original units of the macroeconomic variables using the inverse fitted quantile transformation. Selected variables were then plotted along the PCA path and compared with the proposed density-aware graph path and the realised historical trajectory for each event. The plotted variables were chosen to reflect the macroeconomic quantities most strongly associated with the corresponding historical shock. Since the density-aware graph path consists entirely of observed diffusion-map nodes, its macroeconomic values are taken directly from the corresponding historical observations and require no inverse mapping.

The PCA experiment therefore provides a method-level benchmark between a linear low-dimensional interpolation and the proposed nonlinear, density-aware graph construction. The endpoint reconstruction errors, nearest-neighbour diagnostics and macroeconomic trajectory comparisons are reported in Section 8.

4.11 Robustness design

In this section, we describe the robustness checks we performed to ensure the validity of our results. We hope to see that our qualitative result is stable: density-aware graph paths avoid low-density regions more effectively than straight-line interpolation. Additionally, we want to see a stable difference in the path diagnostics described in Section 4.6 between an ordinary and a density-aware graph path.

4.11.1 Sensitivity to core path parameters

The stress-path construction depends on several tuning parameters introduced in the preceding sections. Although the baseline values are selected using the diagnostics described earlier, it is important to establish that the principal conclusions are not specific to a single parameter configuration. Sensitivity analysis was therefore performed for the diffusion-map bandwidth ε, the number of nearest neighbours k used to construct the graph, and the density-penalty parameter β. In each experiment, one parameter was varied while the remaining parameters were held at their baseline values,

$$\varepsilon = 3, \qquad k = 15, \qquad \beta = 1.5.$$

The analysis was repeated for each of the four start–stress event pairs.

The bandwidth was varied over

$$\varepsilon \in \{1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}.$$

For each value of ε, the diffusion representation was reconstructed and the subsequent density estimation, graph construction and path calculation were repeated using the baseline values of k and β. The purpose of this experiment differs from the bandwidth diagnostics used to select ε in Section 4.3. Those diagnostics assess whether the diffusion operator itself is sufficiently well behaved, whereas the present analysis asks whether the resulting stress-path conclusions remain stable over a reasonable neighbourhood of the selected bandwidth.

Graph sensitivity was assessed using

$$k \in \{5, 10, 15, 20, 25, 30\}.$$

For each value of k, the symmetric nearest-neighbour graph was reconstructed on the fixed baseline diffusion embedding, after which both the unpenalised and density-aware shortest paths were recalculated. This tests whether the conclusions depend strongly on the particular level of graph locality used to approximate movement through the sampled manifold.

For both ε and k, robustness was assessed by comparing the density-aware path directly with the corresponding unpenalised graph path under the same parameter configuration. Let M_{min}, M_{mean}, L_{diff} and D_{turn} denote respectively the minimum path density, arc-length-weighted mean density, latent path length and total turning angle defined in Section 4.6. For a generic metric M, define

$$\Delta M = M^{(\text{aware})} - M^{(\text{ordinary})}.$$

The four reported differences are therefore

$$\Delta M_{\text{min}}, \qquad \Delta M_{\text{mean}}, \qquad \Delta L_{\text{diff}}, \qquad \Delta D_{\text{turn}}.$$

Positive values of the first two quantities indicate that the density-aware path travels through regions of greater historical support than the unpenalised graph path. By contrast, positive values of ΔL_{diff} and ΔD_{turn} indicate the geometric cost associated with this change, in the form of a longer or more strongly turning route. Expressing the results as differences isolates the effect of density awareness from changes to the underlying embedding or graph that affect both path types simultaneously.

The density penalty was varied over

$$\beta \in \{0, 0.25, 0.5, 0.75, 1, 1.5, 2\}.$$

Here the diffusion embedding, density field and k-nearest-neighbour graph were held fixed at their baseline specifications, and only the edge-cost penalty was changed. Since $\beta = 0$ reduces the density-aware objective to the ordinary graph shortest-path problem, a separate unpenalised comparison is unnecessary in this experiment. Instead, the absolute values of

$$M_{\text{min}}, \qquad M_{\text{mean}}, \qquad L_{\text{diff}}, \qquad D_{\text{turn}}$$

were followed as β increased. This provides a direct view of the trade-off between avoiding regions of weak historical support and accepting additional path length or curvature.

No smooth dependence on these parameters is assumed. The stress paths are solutions to a discrete shortest-path problem, and changes in the embedding, graph connectivity or edge weights may cause the optimal route to switch between different sequences of observed states. Robustness is therefore assessed primarily through the persistence of the substantive comparison between density-aware and unpenalised paths across the tested parameter ranges, rather than through smooth variation of the individual metrics. The corresponding sensitivity results are reported in Section 6.5.

4.11.2 Sensitivity to density estimation

The density-aware path construction depends on the estimate of historical support in diffusion space. We therefore test whether the principal empirical conclusion—that density-aware graph paths avoid regions of weak historical support relative to the linear and unpenalised graph benchmarks—is robust to alternative definitions of the density field. In this experiment, the diffusion embedding, nearest-neighbour graph and density-penalty parameter are held fixed at their baseline specifications, while the density estimator used to construct the potential is varied.

The baseline estimator used throughout the main analysis is the graph-degree density introduced in Section 4.4. For a general point $x \in \mathbb{R}^r$ in diffusion space, its unnormalised density estimate is

$$\widehat{\rho}_G(x) = \sum_{j=1}^{N} \exp\left(-\frac{\|x - \psi_j\|^2}{h_{\text{dens}}}\right).$$

As previously, this is rescaled by the mean density at the observed diffusion coordinates,

$$\rho_G(x) = \frac{\widehat{\rho}_G(x)}{N^{-1} \sum_{i=1}^{N} \widehat{\rho}_G(\psi_i)}.$$

To assess sensitivity to this choice, we also consider kernel density estimators (KDEs). For a radial kernel K and bandwidth $b > 0$, the KDE at a general diffusion-space point x takes the form

$$\widehat{\rho}_K(x) = \frac{1}{Nb^r} \sum_{j=1}^{N} K\left(\frac{\|x - \psi_j\|}{b}\right).$$

Three kernel forms are considered. The Gaussian kernel is

$$K_G(u) = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{u^2}{2}\right),$$

the Epanechnikov kernel is

$$K_E(u) \propto \max\{1 - u^2, 0\},$$

and the exponential kernel is

$$K_X(u) \propto \exp(-u).$$

The Gaussian and exponential kernels have infinite support, whereas the Epanechnikov kernel assigns zero weight beyond the bandwidth. The exponential kernel also decays more slowly with distance than the Gaussian kernel. Since each estimated density is subsequently divided by its mean value over the observed diffusion coordinates, multiplicative kernel normalising constants do not affect the resulting normalised density field.

Before using alternative kernels, the Gaussian KDE provides a useful consistency check on the baseline graph-degree estimator. Substituting the Gaussian kernel gives the distance-dependent term

$$\exp\left(-\frac{\|x - \psi_j\|^2}{2b^2}\right).$$

This agrees exactly with the graph-degree kernel when

$$2b^2 = h_{\text{dens}},$$

or equivalently

$$b = \sqrt{\frac{h_{\text{dens}}}{2}}.$$

The remaining factors in the Gaussian KDE are constant with respect to x and disappear under the mean normalisation. The normalised Gaussian KDE should therefore reproduce the graph-degree density when this bandwidth transformation is used. This equivalence is used as an implementation consistency check rather than treating the Gaussian KDE as an additional robustness specification.

The robustness analysis consequently compares the baseline graph-degree estimator with Epanechnikov and exponential KDEs. For both alternative kernels, the bandwidth is fixed at

$$b = \sqrt{\frac{h_{\text{dens}}}{2}},$$

rather than being re-selected separately. This keeps the smoothing parameter fixed and allows the experiment to focus on the effect of changing the kernel form. The different kernels nevertheless have different support and decay properties, so this choice does not imply that they induce identical effective smoothing.

For each density method m, the density is normalised using its own full-sample mean,

$$\rho^{(m)}(x) = \frac{\widehat{\rho}^{(m)}(x)}{N^{-1}\sum_{i=1}^{N}\widehat{\rho}^{(m)}(\psi_i)},$$

and the corresponding potential is constructed as

$$V^{(m)}(x) = -\log\left(\max\{\rho^{(m)}(x), \delta\}\right),$$

where δ is the same density floor introduced in Section 4.4. A new density-aware graph path is then obtained for each estimator by substituting $V^{(m)}$ into the density-weighted edge cost defined in Section 4.5. The unpenalised graph path is unaffected by the density estimator and therefore remains fixed.

The geometry of the linear benchmark is likewise independent of the density estimator. To ensure that changes in its minimum evaluated density cannot arise from changes in discretisation, its number of interpolation points is fixed separately for each event using the baseline graph-degree specification,

$$n_{\text{lin}} = \left\lfloor \frac{n_{\text{ordinary}} + n_{\text{aware,baseline}}}{2} \right\rfloor$$

where n_{ordinary} and $n_{\text{aware,baseline}}$ are the numbers of vertices in the ordinary and baseline density-aware graph paths, respectively. The same linear grid is then used for all density estimators.

For each method, all three path types—linear, unpenalised graph and density-aware graph—are evaluated under the density field associated with that method. At observed graph vertices, density is evaluated using the leave-one-out procedure from Section 4.6. In particular, for an observed node ψ_i, its own kernel contribution is removed while retaining the full-sample normalisation scale. For a KDE this corresponds to

$$\widehat{\rho}^{(m)}_{-i}(\psi_i) = \frac{1}{Nb^r} \sum_{j \neq i} K_m \left(\frac{\|\psi_i - \psi_j\|}{b} \right).$$

Off-sample interior points of the linear path are evaluated against the full historical sample. The observed start and stress endpoints are excluded from the interior-density statistics.

The principal diagnostic is the minimum interior density,

$$M^{(m)}_{\min}(\gamma) = \min_{x \in \gamma^\circ} \rho^{(m)}_{\text{eval}}(x),$$

which is calculated for each event and path type under each density specification. Comparisons are made within a given estimator rather than between the absolute density values produced by different kernels. The relevant robustness question is whether the density-aware path continues to attain greater minimum historical support than the linear and unpenalised graph paths when the definition of density is changed.

Finally, the density-aware graph paths generated by the different estimators are compared visually in diffusion space. This distinguishes robustness of the density ordering from robustness of the path geometry itself: two estimators may support the same qualitative conclusion while assigning different density values or selecting different sequences of historical states. The resulting density comparisons and path geometries are presented in Section 6.5.

4.11.3 HMM robustness

5 Numerical Validation on Toy Model

5.1 Purpose of the validation experiments

5.2 Benchmark manifolds

5.3 Bandwidth and spectral validation

5.4 Recovery of manifold geometry

6 Macroeconomic Manifold and Stress Paths

This is the main results section of the paper

6.1 Empirical embedding results

The diffusion-map construction was applied to the quantile-transformed macroeconomic observations using the density normalisation $\alpha = 1$. The bandwidth and embedding dimension were selected using the diagnostics described in Section 4.3. Since the purpose of this stage was to obtain a suitable geometric representation for the subsequent path analysis, rather than to interpret every individual diffusion coordinate, the results are reported only briefly.

Bandwidth selection. The kernel-mass scaling diagnostic suggested a bandwidth of approximately $\varepsilon = 1.13$. This value was rejected because the corresponding minimum kernel degree was close to one. As the diagonal affinity satisfies $W_{ii} = 1$, this indicated that some observations were almost isolated from the remainder of the sample. The leading diffusion coordinates at small bandwidths were also largely flat, with occasional sharp spikes, rather than displaying stable variation across the sample. Inlcude a plot of this maybe

A bandwidth of

$$\varepsilon = 3$$

was selected instead. At this value, the minimum degree increased to approximately 19.35, and the degree distribution and leading eigenspectrum were stable relative to nearby bandwidths. Increasing the bandwidth further did not materially improve the leading coordinates and would have made the kernel progressively less local. The selected value was therefore the smallest bandwidth that provided a sufficiently connected and spectrally stable diffusion operator.

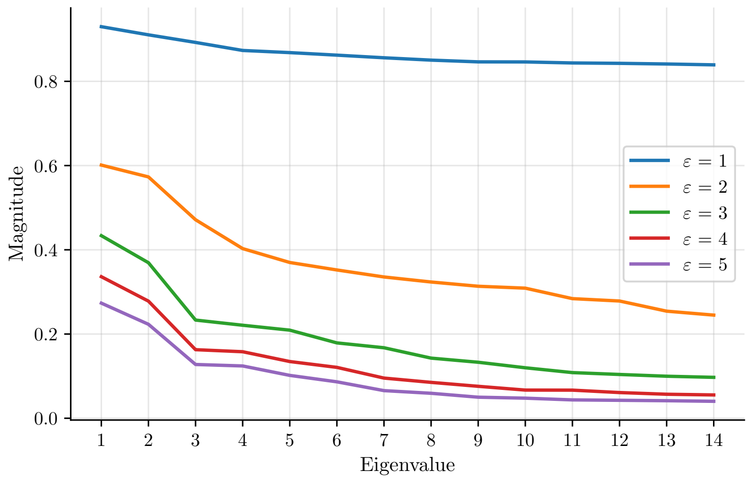


Figure 1: Spectrum of the transition matrix for different bandwidths.

ε	**Min Degree**	**Degree CV**
1	1.20	0.47
2	5.54	0.37
3	19.35	0.28
4	41.77	0.23
5	69.34	0.19

Table 1: Minimum degree and degree CV across bandwidths.

Embedding dimension. Change this section a bit. Add largest spectral gap suggesting $r = 2$. 3d plots suggest $r = 3$, although hard to check for 4d probably; can give this a go plotting the 2,3,4 coordinates and seeing if there is a separation in two different directions like CPI vs UNRATE for coordinates 1,2,3. LNL RMSE and % variance recovered suggest $r = 4$. Near-collision comparison suggests $r = 4$. We choose $r = 4$ because ultimately we are interested in the reconstruction back to macro variables. The leading eigenspectrum did not, by itself, provide an unambiguous cut-off for the embedding dimension. The incremental information supplied by additional coordinates was therefore assessed using the near-collision diagnostic. For each candidate dimension r, pairs of observations that were close in the first r

diffusion coordinates (closest 1%) were identified, subject to a twelve-month temporal exclusion window. The separation of these pairs in the the higher-dimensional space was then measured and percentiles were taken.

Table 2 reports the results. The addition of coordinate $r = 3$ and $r = 4$ both produced meaningful separation of observations that appeared close in the lower-dimensional representation. By contrast, coordinate $r = 5$ provided substantially weaker separation, indicating that the incremental geometric information beyond dimension $r = 4$ was limited. The final embedding dimension was therefore fixed at

$$r = 4.$$

Table 2: Percent of the near-collision pairs that are no longer in the closest x% of pairs in the higher-dimensional embedding. Near-collisions are pairs are the top 1% of closest nodes in the lower-dimensional embedding after applying a twelve-month temporal exclusion window.

Comparison	Near-collision pairs	% Exit 1%	% Exit 5%	% Exit 10%
2D → 3D	2669	58.0	31.1	18.6
3D → 4D	2669	49.5	20.8	10.1
4D → 5D	2669	23.1	1.95	0.26

In Table 2, we can see that adding an extra dimension can add lots of separation. For example, roughly 496 of the near-collision pairs in $r = 2$ are no longer in the top 10% closest pairs in $r = 3$; this clearly shows that there is a considerable amount of new information gained by adding a 3rd coordinate. The results are similar for adding the 4th coordinate; however, there is a dramatic drop-off in the gained separation of near-collision nodes when adding the 5th coordinate.

Structure of the embedding. Figure 2 displays the resulting diffusion coordinates. Observations associated with NBER recessions were visibly displaced from the main concentration of observations in the leading projections. This separation was already present in the first few coordinates, indicating that recessionary periods represent comparatively unusual locations in the learned macroeconomic state space rather than being distinguished only by higher-order coordinates.

The embedding also exhibited systematic variation with NBER labels, inflation rates, and industrial production levels. These patterns support the interpretation of the diffusion coordinates as describing economically meaningful variation in the joint state of the variables. The coordinates themselves are not assigned direct economic labels, however, and are used primarily as a geometric state representation in the subsequent graph and stress-path analysis.

Overall, the selected diffusion map provided a connected and stable low-dimensional representation of the transformed macroeconomic data. This embedding is used as the state space for the graph-based stress paths constructed in the following section.

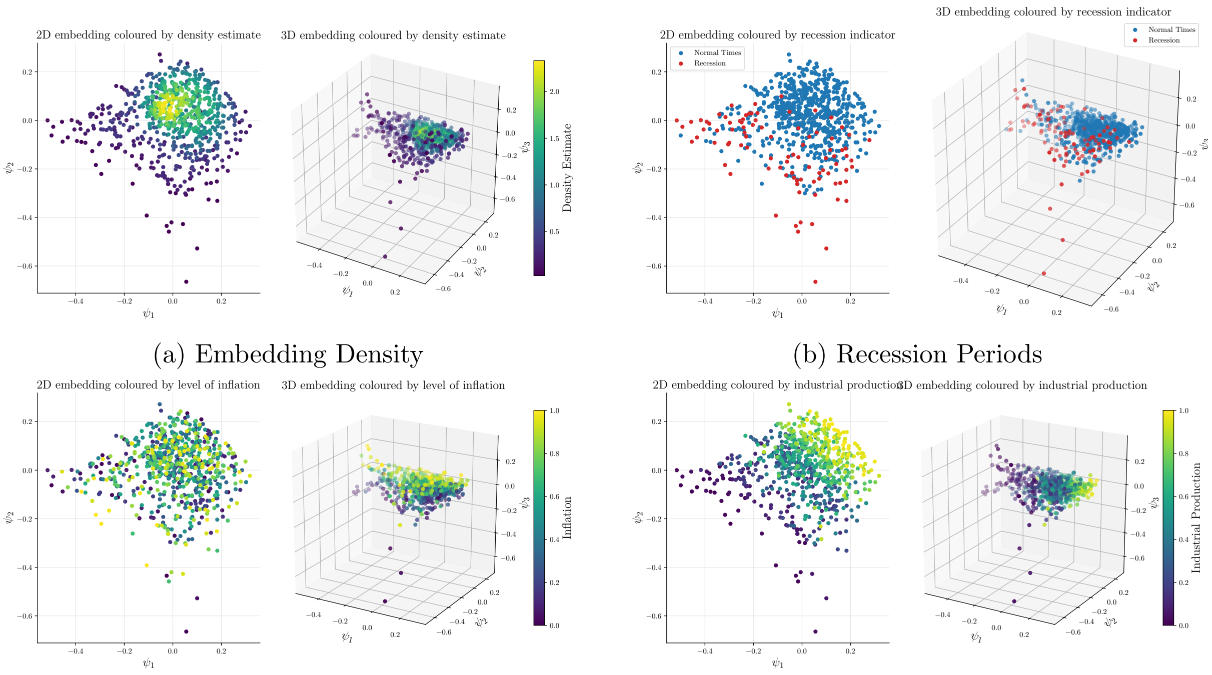
6.2 Embedding density results

6.3 Stress-path results

6.4 Lifted-path results

6.4.1 Choice of lifting method

6.5 Robustness of parameters



(a) Embedding Density

(b) Recession Periods

(c) Inflation

(d) Industrial Production

Figure 2: Diffusion-map representation of the quantile-transformed macroeconomic observations using $\alpha = 1$, $\varepsilon = 3$. While our diffusion dimension is 4-dimensional, here we only show 2D and 3D plots. Observations are coloured by density, NBER recession labels, inflation rates (quantile), and industrial production levels (quantile).

7 Hidden Regimes on the Macroeconomic Manifold

This is the extension

8 PCA benchmark results

9 Conclusions and outlook

References

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Appendices

A First appendix

B Second appendix